MIL-E-1/424B
26 November 1968
SUPERSEDING
MIL-E-1/424A
7 February 1966

MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, RECEIVING

## TYPE 6F4

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, UHF, oscillator

| | | |
|---|---|---|
| Outline<br>Base | } | 4-2 (EIA) Ⓑ |
| Envelope | --- | T4-1/2 |
| Cathode | --- | Coated unipotential |

Base connections:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Element | --- | h | g | a | a | g | h | k |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ebb | Rk | Pp | Ehk | Ik | Alt |
|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Ohms | W | v | mAdc | ft |
| Maximum: | 6.9 | 165 | --- | --- | 2.0 | 90 | 22 | 10,000 |
| Minimum: | 5.7 | --- | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | 6.3 | --- | 150 | 500 (see note 1) | --- | --- | --- | --- |

GENERAL:

Qualification - Required

Ⓑ denotes changes



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | |
| 1256 | Electrode current (1) (anode) | Ebb = 80 Vdc; Rk = 150 ohms | 0.65 | II | Ib | 8 | 19.5 | mAdc |
| 1256 | Electrode current (2) (anode) | Ec = -15 Vdc; Ebb = 80 Vdc; Rk = 150 ohms | 0.65 | II | Ib | --- | 300 | µAdc |
| 1231 | Emission | Eb = Ec = 10 Vdc; Rk = 0 (see note 4) | 0.65 | II | Is | 40 | --- | mAdc |
| 1266 | Total grid current | See note 4 | 0.65 | II | Ic | 0 | -2.0 | µAdc |
| 1301 | Heater current | | 0.65 | II | If | 210 | 240 | mA |
| 1306 | Transconductance | Ebb = 80 Vdc; Rk = 150 ohms | 0.65 | II | Sm | 4,000 | 8,000 | µmhos |
| 1201 | Short and discontinuity detection | See note 3 | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1031 | Low-frequency vibration | Ec1 = -5.0 Vdc; Rk = 0; Rp = 10,000 ohms | 6.5 | S3 | Ep | --- | 300 | mVac |
| (B) 1336 | Heater-cathode leakage | | 6.5 | S3 | Ihk | --- | 20 | µAdc |
| 1316 | Amplification factor | Ebb = 80 Vdc; Rk = 150 ohms | 6.5 | S3 | Mu | 12 | 22 | --- |
| 1331 | Direct-interelectrode capacitance | No shield | 6.5 | Code E | Cgp<br>Cin<br>Cout | 1.5<br>1.4<br>0.46 | 2.1<br>2.4<br>0.68 | pF<br>pF<br>pF |
| (B) 1027 | Temperature cycling | See note 2 | 4.0 | I | --- | --- | --- | --- |
| (B) 1211 | Insulation of electrodes | | 4.0 | S3 | R | 10 | --- | Meg |
| 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| (B) 1501 | Intermittent life | Group A; Ehk = 90; Rg = 0.1 Meg | --- | --- | --- | --- | --- | --- |
| (B) --- | Life-test end point (intermittent) (250 hours) | Transconductance | --- | --- | Sm | 3,000 | --- | µmhos |



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NOTES:

1. The reference point for all potentials shall be the negative terminal of the cathode resistor.

2. The representative sample consisting of each day's production shall be subjected to the following test: (B)

   The tubes shall be removed from boiling water after a minimum period of 2 minutes at 100° C, and immediately (within 30 seconds) placed in an atmosphere not warmer than -40° C for at least 10 minutes, after which time the outer surface of the bulb shall have attained a temperature of -10° C, or colder. The tubes shall then be returned to room temperature and shall be subjected to a mechanical pin loading test as follows:

   The pins shall rest on a cylinder of 29/32-inch (23.02 mm) inside diameter. A weight of at least 16 pounds shall be gradually applied to the top or bottom of the tube in a direction perpendicular to the plane of the lead wires. The tube shall then be inverted and the weight applied from the opposite direction.

   At the conclusion of the above thermal cycling and mechanical tests, the tubes shall be held for a period of at least 24 hours. At the end of this period, the tubes shall be tested for air leakage, as indicated by failure to pass the grid-current test.

3. Testing shall comply with method 1201 except the test equipment shall be ac operated and capable of indicating tap shorts of 50,000 ohms, or less. The tapping shall be performed manually with a felt tapper.

4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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